D.M.E.S. DEWAGEDARA
E/21/087

1)

```
mysql> CREATE TABLE Student(
    -> Name VARCHAR(50),
    -> RegNO INT PRIMARY KEY,
    -> GPA DECIMAL(3,2),
    -> ConvocationRegNo INT,
    -> Class VARCHAR(50));
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> INSERT INTO Student (Name, RegNo, GPA)
    -> VALUES
    -> ('Akila', 154, 3.80),
    -> ('Ishta', 149, 3.55),
    -> ('Chamara', 327, 3.25),
    -> ('Haritha', 118, 3.75),
    -> ('Saadia', 147, 4.00),
    -> ('Thamish', 379, 3.95),
    -> ('Nimuthu', 398, 3.25),
    -> ('Ridma', 155, 3.85),
    -> ('Ravindu', 224, 3.10),
    -> ('Vilakshan', 373, 3.70);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Student;
+-----------+-------+------+-----------------+-------+
| Name      | RegNO | GPA  | ConvocationRegNo | Class |
+-----------+-------+------+-----------------+-------+
| Haritha   |   118 | 3.75 |            NULL | NULL  |
| Saadia    |   147 | 4.00 |            NULL | NULL  |
| Ishta     |   149 | 3.55 |            NULL | NULL  |
| Akila     |   154 | 3.80 |            NULL | NULL  |
| Ridma     |   155 | 3.85 |            NULL | NULL  |
| Ravindu   |   224 | 3.10 |            NULL | NULL  |
| Chamara   |   327 | 3.25 |            NULL | NULL  |
| Vilakshan |   373 | 3.70 |            NULL | NULL  |
| Thamish   |   379 | 3.95 |            NULL | NULL  |
| Nimuthu   |   398 | 3.25 |            NULL | NULL  |
+-----------+-------+------+-----------------+-------+
```

2)

```
mysql> CREATE TABLE Convocation (
    -> ConvocationRegNo INT PRIMARY KEY,
    -> RegNo INT,
    -> LastName VARCHAR(50),
    -> Address VARCHAR(100),
    -> Age INT,
    -> FOREIGN KEY (RegNo) REFERENCES Student(RegNo));
Query OK, 0 rows affected (0.05 sec)
```

3)

```
mysql>
mysql> CREATE PROCEDURE RegisterStudent(
    ->     IN insertConvocationRegNo INT,
    ->     IN insertRegNO INT,
    ->     IN insertLastName VARCHAR(50),
    ->     IN insertAddress VARCHAR(100),
    ->     IN insertAge INT
    -> )
    -> BEGIN
    ->     INSERT INTO Convocation (ConvocationRegNo, RegNo, LastName, Address, Age)
    ->     VALUES (insertConvocationRegNo, insertRegNO, insertLastName, insertAddress, insertAge);
    ->
    ->     UPDATE Student
    ->     SET ConvocationRegNo = insertConvocationRegNo
    ->     WHERE RegNo = insertRegNO;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

4)

```
mysql> CALL RegisterStudent(1, 154, 'Perera', 'Colombo', 24);
Query OK, 1 row affected (0.02 sec)

mysql> CALL RegisterStudent(2, 118, 'Silva', 'Kandy', 23);
Query OK, 1 row affected (0.01 sec)

mysql> CALL RegisterStudent(3, 147, 'Khan', 'Jaffna', 22);
Query OK, 1 row affected (0.01 sec)

mysql> CALL RegisterStudent(4, 379, 'Fernando', 'Galle', 24);
Query OK, 1 row affected (0.01 sec)

mysql> CALL RegisterStudent(5, 155, 'Senanayake', 'Kurunegala', 23);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM Convocation;
+-----------------+-------+------------+------------+------+
| ConvocationRegNo | RegNo | LastName   | Address    | Age  |
+-----------------+-------+------------+------------+------+
|               1 |   154 | Perera     | Colombo    |   24 |
|               2 |   118 | Silva      | Kandy      |   23 |
|               3 |   147 | Khan       | Jaffna     |   22 |
|               4 |   379 | Fernando   | Galle      |   24 |
|               5 |   155 | Senanayake | Kurunegala |   23 |
+-----------------+-------+------------+------------+------+
5 rows in set (0.00 sec)
```

5)

```
mysql> CREATE VIEW Registered AS
    -> SELECT s.Name,s.RegNo,s.GPA,s.ConvocationRegNo,c.Address,c.Age
    -> FROM Student s
    -> JOIN Convocation c ON s.RegNO=c.RegNo;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Registered;
+---------+-------+------+-----------------+------------+------+
| Name    | RegNo | GPA  | ConvocationRegNo | Address    | Age  |
+---------+-------+------+-----------------+------------+------+
| Akila   |   154 | 3.80 |               1 | Colombo    |   24 |
| Haritha |   118 | 3.75 |               2 | Kandy      |   23 |
| Saadia  |   147 | 4.00 |               3 | Jaffna     |   22 |
| Thamish |   379 | 3.95 |               4 | Galle      |   24 |
| Ridma   |   155 | 3.85 |               5 | Kurunegala |   23 |
+---------+-------+------+-----------------+------------+------+
5 rows in set (0.00 sec)
```

6)

```
mysql> CREATE VIEW NotRegistered AS
    -> SELECT s.Name,s.RegNo,s.GPA,'' AS Address,NULL AS Age
    -> FROM Student s
    -> WHERE s.ConvocationRegNo IS NULL;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM NotRegistered;
+-----------+-------+------+---------+------+
| Name      | RegNo | GPA  | Address | Age  |
+-----------+-------+------+---------+------+
| Ishta     |   149 | 3.55 |         | NULL |
| Ravindu   |   224 | 3.10 |         | NULL |
| Chamara   |   327 | 3.25 |         | NULL |
| Vilakshan |   373 | 3.70 |         | NULL |
| Nimuthu   |   398 | 3.25 |         | NULL |
+-----------+-------+------+---------+------+
5 rows in set (0.00 sec)
```

7)

```
mysql>
mysql> CREATE TABLE LateRegistration(
    -> ConvocationRegNo INT PRIMARY KEY,
    -> RegNo INT,
    -> LastName VARCHAR(50),
    -> Address VARCHAR(100),
    -> Age INT,
    -> FOREIGN KEY(RegNo) REFERENCES Student(RegNo));
Query OK, 0 rows affected (0.05 sec)
```

8)

```
mysql> DELIMITER //
mysql> CREATE TRIGGER AfterRegistration
    -> AFTER INSERT ON LateRegistration
    -> FOR EACH ROW
    -> BEGIN
    ->  UPDATE Student
    ->  SET ConvocationRegNo=NEW.ConvocationRegNo
    ->  WHERE RegNo = NEW.RegNo;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER;
    ->
```

9)

```
mysql> INSERT INTO LateRegistration VALUES (6, 149, 'Perera', 'Matara', 22);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO LateRegistration VALUES (7, 327, 'Rajapaksha', 'Anuradhapura', 25);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO LateRegistration VALUES (8, 398, 'Karunaratne', 'Badulla', 23);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO LateRegistration VALUES (9, 224, 'Ekanayake', 'Colombo', 24);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO LateRegistration VALUES (10, 373, 'Pathirana', 'Galle', 24);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM  LateRegistration;
+------------------+-------+-------------+--------------+------+
| ConvocationRegNo | RegNo | LastName    | Address      | Age  |
+------------------+-------+-------------+--------------+------+
|                6 |   149 | Perera      | Matara       |   22 |
|                7 |   327 | Rajapaksha  | Anuradhapura |   25 |
|                8 |   398 | Karunaratne | Badulla      |   23 |
|                9 |   224 | Ekanayake   | Colombo      |   24 |
|               10 |   373 | Pathirana   | Galle        |   24 |
+------------------+-------+-------------+--------------+------+
5 rows in set (0.00 sec)
```

10)

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE AssignClass()
    -> BEGIN
    ->     UPDATE Student
    ->     SET Class =
    ->         CASE
    ->             WHEN GPA >= 3.7 THEN 'First Class Honors'
    ->             WHEN GPA >= 3.3 THEN 'Second Class Honors - Upper Division'
    ->             WHEN GPA >= 2.7 THEN 'Second Class Honors - Lower Division'
    ->             WHEN GPA >= 2.0 THEN 'Third Class Honors'
    ->             ELSE 'No Honors'
    ->         END;
    -> END //
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL AssignClass();
Query OK, 10 rows affected (0.03 sec)
```